

MAT 426, Class 3 Homework

(Due Tuesday, February 17, 2026)

1. Suppose L is an $n \times n$ lower triangular matrix and $\mathbf{b}$ is an $n \times 1$ vector.
 - (a) How many floating point operations (flops) are needed to solve $L\mathbf{x} = \mathbf{b}$?
 - (b) Does the order of magnitude change if L is unit lower triangular?
 - (c) Does the number of flops change if we solve $U\mathbf{x} = \mathbf{b}$ where U is an $n \times n$ upper triangular matrix?
 - (d) How many floating point operations are needed to solve $A\mathbf{x} = \mathbf{b}$ using LU decomposition?
Recall from Class 3 that the total number of flops for LU decomposition is $\frac{2n^3}{3} + \mathcal{O}(n^2)$
2. Suppose A is an $n \times n$ invertible matrix and $\mathbf{b}$ an $n \times 1$ vector.
 - (a) How many row operations are needed to reduce the matrix $[A|I]$ to $[I|A^{-1}]$?
 - (b) How many row operations are needed to compute $\mathbf{x} = A^{-1}\mathbf{b}$?